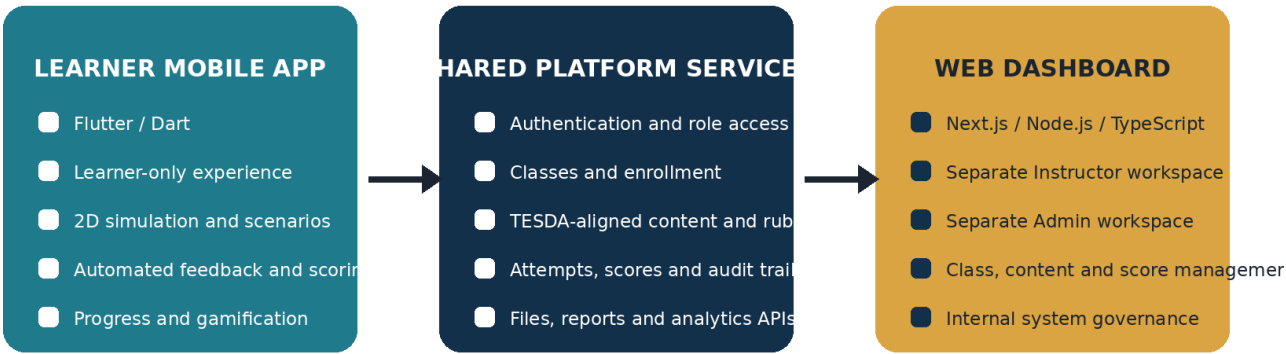


PRODUCT REQUIREMENTS DOCUMENT

Gamified Simulation Platform with Automated Skill Evaluation for NC II Computer Systems Servicing

Combined implementation baseline for the Learner Mobile App and the Admin/Instructor Web Dashboard

Two-Product Platform Boundary



One shared source of truth; separate user experiences and permissions.

Mobile application	Flutter and Dart
Web dashboard	Next.js, Node.js and TypeScript
Document version	1.0 - Scope and implementation baseline
Prepared	07 August 2026

Purpose: provide a single, controlled reference for identifying what is already implemented, partially implemented, or still missing in the existing codebases.

Document Control

Field	Value
Document status	Approved baseline for code gap assessment and capstone implementation planning
Primary source	Uploaded Chapter 1 and Chapter 2 draft, especially the Objectives of the Study and Scope and Limitation sections
Secondary source	Fourteen outline-defense recommendations supplied by the proponents
Implementation source of truth	Existing mobile code in Flutter/Dart and existing web dashboard code in Next.js, Node.js and TypeScript
What this document does not claim	It does not claim which features are already complete because the source code was not provided for audit in this task.
How to use this PRD	Compare each requirement and acceptance criterion against the current code; mark Implemented, Partial, Missing or Not Applicable in the audit checklist.

Important documentation alignment issue

The uploaded Chapter 2 draft refers to Unity and Firebase, while the current implementation was confirmed as Flutter/Dart for mobile and Next.js/Node.js/TypeScript for the web dashboard. This PRD uses the current implementation stack as the engineering baseline. The research manuscript should later be revised so its technology section, diagrams and deployment plan match the actual system.

Contents

1. Executive Summary
2. Review of the Study Objectives
3. Product Vision and Scope Strategy
4. Product and Role Boundaries
5. Learner Mobile App Requirements
6. Instructor Web Dashboard Requirements
7. Admin Web Dashboard Requirements
8. TESDA-Aligned Evaluation, Scoring and COC Rules
9. Learning Content and Interaction Requirements
10. Shared Platform and Data Requirements
11. Permission Matrix
12. Core Workflows
13. Non-Functional Requirements
14. Scope Limitations and Explicitly Out-of-Scope Items
15. Priority and Release Baseline
16. Acceptance and Definition of Done
17. Development Gap-Audit Checklist
- Appendix A. Panel Recommendation Traceability
- Appendix B. Open Decisions
- Appendix C. Glossary

1. Executive Summary

Product problem. CSS NC II learners need more opportunities to practice technical procedures, receive immediate feedback and prepare for competency-based assessment despite limited laboratory equipment and practice time. The proposed product supplements, but does not replace, physical laboratory training.

Product solution. A two-product platform will provide a learner-facing mobile application for 2D scenario-based practice and a role-based web dashboard with clearly separated Instructor and Admin workspaces. Both products use one shared backend and data model so classes, modules, attempts, scores, progress and reports remain synchronized.

Primary scope decision. Learners use the mobile application. Instructors and administrators use the web dashboard. This resolves the panel recommendation that instructors must have a separate system while keeping system management and instructional work distinct.

PRD scope rule

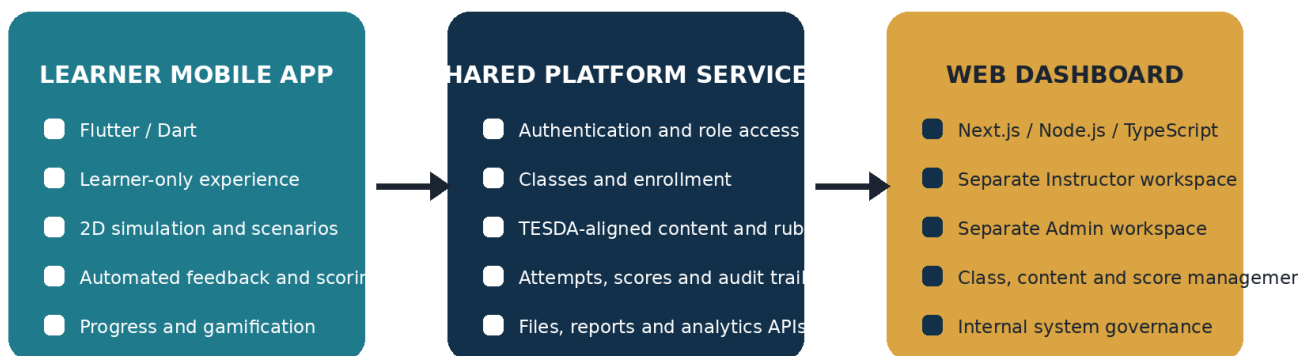
Every feature in this document must support at least one study objective, one panel recommendation, or a necessary shared-platform requirement. Features that do not satisfy that rule are deferred to prevent scope creep.

1.1 Product Outcomes

- **Learner practice:** Learners can complete structured 2D CSS NC II activities repeatedly in a safe environment.
- **Competency alignment:** Modules, criteria, task weights and final scores reference an approved TESDA booklet or rubric version.
- **Objective evaluation:** The system records learner actions and applies transparent rule-based and score-based logic.
- **Instructor control:** Instructors manage classes, content, attempts and final scores without receiving system-administration privileges.
- **Admin governance:** Administrators manage accounts, access, configuration, audit records and internal system concerns.
- **Actionable analytics:** Learners and instructors can view progress, competency results, strengths and areas for improvement.

1.2 Product Structure

Two-Product Platform Boundary



One shared source of truth; separate user experiences and permissions.

Figure 1. Product boundary used by this PRD.

2. Review of the Study Objectives

Source review. The uploaded study defines one primary objective and five specific objectives: scenario-based 2D simulation, gamification, TESDA-aligned competency modules, automated skill evaluation and real-time performance analytics. Its scope limits the system to predefined 2D scenarios, selected DNSC ICT learners, supplementary use, non-advanced hardware and no official TESDA certification authority.

Study Objective	PRD Interpretation	Primary Product Area
Objective 1: Scenario-based 2D simulation modules	The mobile app must contain structured scenarios and interactive tasks, not static lessons only.	Mobile app; instructor content management
Objective 2: Gamification	The system must provide points, levels/rewards/achievements and progress indicators, but these must not distort competency scoring.	Mobile app; shared scoring services
Objective 3: TESDA-aligned competency modules	Every module and criterion must be mapped to an approved competency/COC structure and versioned source.	Shared content model; instructor and admin dashboard
Objective 4: Automated skill evaluation	The system must capture learner actions, apply predefined rules, calculate provisional results and retain criterion-level evidence.	Mobile app event capture; evaluation service; instructor review
Objective 5: Performance analytics dashboard	Learners need personal progress views; instructors need class and student analytics; admins need system-level oversight.	Mobile progress screens; web dashboards

2.1 Key Findings from the Objective Review

1. The product cannot satisfy the study by providing quizzes alone; practical, scenario-based interaction is a core requirement.
2. Automated scoring must be explainable. A learner or instructor should be able to see which criterion was met, missed or adjusted.
3. TESDA alignment is a hard content and scoring constraint, but the platform itself is not an official TESDA assessment or certification tool.
4. The analytics objective requires shared data between mobile and web; separate disconnected databases would fail the intended monitoring workflow.
5. The panel recommendations expand the original Instructor/Admin actor into two distinct permission domains and add LMS-lite class management.

3. Product Vision and Scope Strategy

3.1 Vision Statement

Provide CSS NC II learners with an accessible, interactive and competency-oriented practice environment, while giving instructors reliable tools to guide learning and giving administrators secure control over system access and operations.

3.2 Product Principles

Principle	Meaning
TESDA first	The approved booklet, competency standards and instructor-validated rubric are the content and scoring source of truth.

Principle	Meaning
Practice is not certification	The platform prepares and supplements learners; it does not issue an official NC II result.
Separate duties	Instructional actions belong to instructors; internal platform governance belongs to administrators.
Evidence before score	Every automated or manually adjusted result must be traceable to attempt evidence and criteria.
Interactive, not decorative	Game elements and visuals must support learning; oversized logos and unnecessary decoration are minimized.
Build the capstone core first	AI and advanced enhancements remain optional until the required objectives and panel items are complete.

3.3 Scope Strategy

- **P0 - Mandatory capstone scope:** All five study objectives, role separation, class management, score review, COC bypass, TESDA-based rules, required UI corrections and core reporting.
- **P1 - Conditional enhancement:** Instructor-assisted generative AI for quiz drafting, richer scenario media and additional analytics after P0 is stable.
- **P2 - Future research or expansion:** 3D/AR/VR, cross-institution deployment, advanced hardware configurations, adaptive machine learning and official external integrations.

4. Product and Role Boundaries

4.1 Roles

Role	Boundary
Learner / Student	Uses the Flutter mobile application to access assigned classes and modules, perform scenarios, receive feedback, earn approved points and view personal progress.
Instructor	Uses a separate web workspace to manage classes, learning content, files, learner access, COC sequencing, attempts, final scores and instructional reports.
Administrator	Uses a separate web workspace to manage accounts, roles, system access, configuration, content governance, storage, audit logs and internal system problems.
System services	Shared authentication, APIs, evaluation engine, data storage, file storage, reporting and audit mechanisms used by both products.

4.2 Internal vs. External Problem Ownership

Ownership	Examples
Internal system problems - Admin	Account creation/removal; role assignment; login/access failures; system configuration; global rubric version; database/storage concerns; audit and security issues; platform availability; data recovery.
External learning problems - Instructor	Class setup; student enrollment; activity availability; learner confusion; module sequencing; COC bypass; score review; incorrect answer placement; final grading; learning files; intervention based on performance.
Learner self-service	Profile viewing, assigned activity access, attempt submission, feedback, retry within instructor rules and progress viewing. Learners cannot alter scores, roles or access rules.

4.3 Shared Platform Boundary

- The mobile app and web dashboard must use the same identity, class, module, competency, attempt, score and progress records.
- Role checks must be enforced by the backend, not only hidden in the user interface.
- Instructor and Admin workspaces may exist in one Next.js codebase, but their routes, navigation and permissions must be clearly separated.
- A database or authentication provider is not prescribed here because the current backend implementation was not supplied. The existing provider should be retained unless the code audit proves it cannot meet these requirements.

5. Learner Mobile App Requirements

Product purpose. The Flutter mobile application is the learner-facing simulation and practice product. It must remain focused on learning activities and personal progress rather than administrative functions.

ID	Requirement	Priority	Acceptance Summary
MOB-001	Authenticate learners and validate that the account is active and authorized for the Learner role. Deactivated accounts must be denied access while records remain preserved.	P0	Role and account-state tests pass.
MOB-002	Display the learner profile, enrolled classes, instructor, assigned modules and availability state.	P0	Assignments match web dashboard data.
MOB-003	Present modules in the configured COC/competency sequence, including locked, available, in-progress and completed states.	P0	Sequence rules are enforced consistently.
MOB-004	Allow a module to contain text-based and/or video-based scenarios before an activity begins.	P0	Both supported content types render correctly.

ID	Requirement	Priority	Acceptance Summary
MOB-005	Support interaction types beyond drag-and-drop, including sequencing, hotspot/component selection, multiple choice, tool selection and branching troubleshooting decisions.	P0	At least three interaction types are demonstrated.
MOB-006	In assessment mode, remove visible answer boxes for drag-and-drop. Valid placement may use subtle feedback only after an action or according to the approved activity design.	P0	No permanent target boxes are visible during assessment.
MOB-007	Provide full-screen and zoom controls suitable for classroom demonstration and accessibility.	P0	Content remains usable at supported zoom levels.
MOB-008	Record attempt start, completion, selected actions, action order, errors, elapsed time, criteria matched and submission state.	P0	Attempt evidence is stored and retrievable.
MOB-009	Receive automated criterion-level feedback and a provisional result from the rule-based evaluation service.	P0	Result matches configured rules and weights.
MOB-010	Show the learner final score only after the configured release rule; clearly distinguish provisional and instructor-finalized results.	P0	Status labels prevent score ambiguity.
MOB-011	Display competency progress, module completion, attempts, achievements and approved gamification points.	P0	Personal analytics reflect stored records.
MOB-012	Apply instructor-defined retry limits and availability windows without deleting previous attempts.	P0	Attempt history remains intact.
MOB-013	Provide safe error handling when the network, content or scoring service is unavailable; prevent duplicate submissions.	P0	Retries do not create duplicate final attempts.
MOB-014	Use a balanced visual layout with minimized logos/figures and readable controls on supported devices.	P0	Usability review confirms visual balance.
MOB-015	Keep AI generation out of the learner experience; learners only receive instructor-reviewed and published content.	P1	No unreviewed AI content is presented.

5.1 Required Mobile Screens

Screen Area	Minimum Content
Authentication	Login, account state message and secure logout.
Home / Classes	Enrolled classes, assigned activities, due/availability state and progress summary.
Module / COC Map	TESDA-aligned module order, lock state, bypass state and completion status.
Scenario Brief	Text/video context, objectives, instructions, safety notes and start action.
Simulation / Activity	Interactive 2D activity with zoom/full-screen, feedback and assessment controls.
Result and Feedback	Provisional/final label, criterion breakdown, mistakes, correct steps and retry state.
Progress and Achievements	Competency progress, scores, attempts, points, badges/rewards and completed modules.
Profile	Learner details, class memberships and account status; no role or score editing.

5.2 Mobile-Specific Limitations

- No instructor or administrator management functions are included in the learner mobile app.
- No official NC II certificate, assessment result or assessor authority is generated.
- No advanced hardware emulation, electrical safety simulation, 3D laboratory or real device control is included.
- New mobile operating-system targets are not added solely for this PRD; the supported targets follow the existing Flutter project and capstone deployment plan.

6. Instructor Web Dashboard Requirements

Product purpose. The Instructor workspace handles external learning and assessment concerns. It operates like a focused LMS for classes, CSS NC II content, learner monitoring and final score review, without granting internal system-administration authority.

ID	Requirement	Priority	Acceptance Summary
INS-001	Provide a dedicated Instructor login destination, navigation and route guard separate from the Admin workspace.	P0	Instructor cannot access Admin-only routes or APIs.
INS-002	Create, edit, archive and view classes; define class name, section, term, schedule metadata and assigned modules.	P0	Class lifecycle works without deleting historical records.
INS-003	Enroll learners into classes, remove class membership where permitted, and view enrollment history.	P0	Membership changes synchronize to mobile.

ID	Requirement	Priority	Acceptance Summary
INS-004	Deactivate an assigned learner account with a required reason, while preserving attempts, scores and records. The instructor cannot permanently delete an account.	P0	Deactivated learner cannot log in; data remains.
INS-005	Create, edit, preview, publish, unpublish and archive learning modules and scenarios mapped to competencies/COCs.	P0	Only published content is learner-visible.
INS-006	Upload and manage learning files such as PDFs, images, videos and supporting documents subject to configured limits.	P0	Files are access-controlled and downloadable/streamable.
INS-007	Create and manage quizzes, answer keys, criteria and task weights based on the approved TESDA booklet or instructor-validated rubric.	P0	Every published quiz has a rubric/version reference.
INS-008	Optionally generate draft quiz items using generative AI. AI output must remain a draft until the instructor reviews, edits and explicitly publishes it.	P1	No AI-generated item auto-publishes.
INS-009	Configure module order, prerequisites and availability for a class.	P0	Learner access matches class configuration.
INS-010	Bypass COC 1-4 sequencing for a learner or class when authorized, with scope, reason, date and instructor identity recorded. Bypass unlocks access but does not automatically mark competency as passed.	P0	Audit record exists and no automatic passing occurs.
INS-011	Monitor class and learner progress in real time or near-real time, including module status, attempts, scores, competency achievements and common errors.	P0	Dashboard matches attempt and score records.
INS-012	Open attempt details to review learner actions, sequence, responses, automated criteria and feedback.	P0	Evidence is available before finalization.
INS-013	Confirm or edit the automated score, record the final score, add remarks and require a reason for every adjustment.	P0	Original and adjusted values remain auditable.
INS-014	Release or withhold finalized results to learners according to class policy.	P0	Mobile displays only released results.

ID	Requirement	Priority	Acceptance Summary
INS-015	Generate class, learner, module and competency reports and export approved report formats.	P0	Report totals reconcile with stored results.
INS-016	View intervention indicators such as repeated errors, low competency areas and incomplete modules.	P0	Indicators link to supporting learner data.
INS-017	Access only assigned classes and authorized learners unless broader scope is explicitly granted by Admin.	P0	Authorization tests prevent cross-class access.

6.1 Required Instructor Areas

Workspace Area	Minimum Content
Dashboard	Class overview, recent attempts, pending score reviews, alerts and competency trends.
Classes	Class creation, module assignment, enrollment, learner status and class configuration.
Content	COCs/competencies, modules, scenarios, activities, quizzes, criteria, files and publishing workflow.
Assessments	Attempt evidence, automated result, score adjustment, remarks, finalization and release.
Analytics and Reports	Student/class/module progress, common errors, competency achievement and export.
Audit View	Instructor actions affecting access, bypasses, content and scores.

7. Admin Web Dashboard Requirements

Product purpose. The Admin workspace handles internal system concerns, access governance and platform integrity. Admins do not replace instructors in routine teaching or score finalization.

ID	Requirement	Priority	Acceptance Summary
ADM-001	Provide a dedicated Admin workspace, navigation and route guard separate from the Instructor workspace.	P0	Admin access is explicit and audited.
ADM-002	Create and manage administrator, instructor and learner accounts; assign roles and initial access state.	P0	Role changes take effect across products.

ID	Requirement	Priority	Acceptance Summary
ADM-003	Activate, deactivate, restore and permanently remove instructor or learner accounts according to policy. Permanent removal is Admin-only and must protect required historical records.	P0	Deletion follows retention safeguards.
ADM-004	Manage instructor scope, assigned classes and organization-level permissions.	P0	Instructor data access follows assignment.
ADM-005	Resolve internal account and access problems, including locked accounts, invalid role assignments and system-level availability concerns.	P0	Support action is logged.
ADM-006	Manage global system settings, file limits, allowed content types, report configuration and feature flags.	P0	Configuration changes are versioned/audited.
ADM-007	Register and activate approved TESDA booklet/rubric versions used by modules and scoring criteria. Exact values must be entered from the approved source rather than guessed.	P0	Only active approved versions can be published.
ADM-008	View system-wide classes, users, modules, attempts, storage usage and health indicators without automatically receiving instructor editing rights.	P0	Read scope is broader than write scope.
ADM-009	Review audit logs for logins, role changes, deactivations, deletions, COC bypasses, content publication and score adjustments.	P0	Critical actions are traceable.
ADM-010	Manage backups, recovery procedures and data export required for capstone continuity and demonstration.	P0	Restore procedure is documented and tested.
ADM-011	Manage global content availability or remove prohibited/corrupt content while preserving an audit record.	P0	Removed content is not silently erased.
ADM-012	View system-level analytics and generate administrative reports.	P0	System totals reconcile with source records.
ADM-013	Avoid routine alteration of learner final scores. Any emergency administrative correction requires a distinct privilege, reason and audit entry.	P0	Admin score changes are exceptional and traceable.

ID	Requirement	Priority	Acceptance Summary
ADM-014	Manage service configuration for optional AI quiz generation without allowing AI to publish content or grade learners autonomously.	P1	AI configuration is access-controlled.

8. TESDA-Aligned Evaluation, Scoring and COC Rules

Source-of-truth rule

The TESDA booklet or training regulation approved by the study team and instructor is the basis for competency criteria, sequence, task weights, pass conditions and scoring. This PRD intentionally does not invent exact percentages or thresholds that were not supplied.

Competency Evaluation and Score Finalization Flow

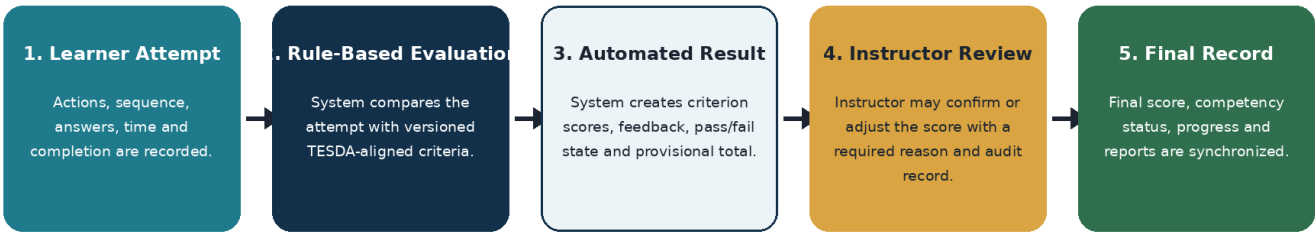


Figure 2. Required evaluation and finalization flow.

8.1 Scoring Model

Record	Requirement
Competency score	Assessment result calculated from approved task criteria and weights. Used for learner performance, pass/fail and instructor finalization.
Gamification points	Engagement/reward value derived from approved completion or performance rules. It must be clearly labeled and must not replace the competency score.
Automated/provisional score	System-calculated result before instructor confirmation where review is required.
Final score	Instructor-confirmed or instructor-adjusted score, with final status, remarks and release state.
Criterion evidence	Actions, answers, sequence and completion facts used to explain how each criterion was evaluated.

8.2 Mandatory Evaluation Rules

ID	Rule
EVAL-001	Every assessable activity references a competency/COC, rubric version and one or more criteria.
EVAL-002	Rules must be deterministic and testable for predefined procedures.
EVAL-003	Correct, incorrect, incomplete and out-of-sequence actions are recorded according to configured criteria.
EVAL-004	The automated engine returns criterion-level results, not only a total score.
EVAL-005	An instructor score adjustment stores original score, new score, reason, remarks, instructor, timestamp and affected criteria.
EVAL-006	Finalized scores are immutable to ordinary users; corrections create a new audited revision rather than silently overwriting history.
EVAL-007	COC bypass unlocks access only. It must not award points, completion or competency without a recorded assessment or instructor decision.
EVAL-008	A rubric version used by a submitted attempt remains attached to that attempt even if a newer rubric is later activated.
EVAL-009	Learners cannot edit attempts, scores, criteria, bypasses or final status.
EVAL-010	The platform must display that results are for supplementary learning and are not an official TESDA certification decision.

9. Learning Content and Interaction Requirements

9.1 Content Model

Content Object	Definition
Competency / COC	Top-level TESDA-aligned learning and assessment grouping.
Module	A structured set of lessons, scenarios and activities aligned to one or more competencies.
Scenario	Context presented through text, video or both; includes objective, constraints and expected outcome.
Activity	Interactive practice or assessment such as sequence, selection, drag-and-drop, configuration or troubleshooting.
Criterion	Observable rule used to evaluate a learner action or result.
Resource file	Instructor-managed supporting document, image, video or reference material.
Quiz item	Instructor-created or instructor-approved item with answer key, explanation and rubric mapping.

9.2 Interaction and UI Rules from the Panel

ID	Rule
UI-001	Provide zoom and full-screen controls for actual demonstration and learner accessibility.
UI-002	Remove permanent boxes/targets from drag-and-drop assessment screens. Practice mode may use optional guided hints if clearly distinguished.
UI-003	Use more than drag-and-drop. Each major competency should include appropriate decision, selection, sequencing or troubleshooting interaction.
UI-004	Support text and video scenarios as learning context.
UI-005	Minimize oversized logos, figures and decorative assets. Learning content and controls take visual priority.
UI-006	Use clear state labels: practice vs assessment, provisional vs final score, locked vs bypassed vs completed.
UI-007	Use readable contrast, scalable text, keyboard-accessible dashboard controls and touch targets suitable for mobile.
UI-008	Prevent accidental submission or score finalization through confirmation and recoverable drafts where appropriate.

9.3 Generative AI Boundary

- AI is optional and P1 because the panel said it should be included only if suitable.
- AI may draft quiz items, distractors, explanations or scenario ideas for an instructor.
- The instructor must review, edit, map to the correct competency/rubric and publish the item manually.
- AI must not autonomously determine official competency, edit final scores, publish directly to learners or replace the rule-based evaluation engine.
- Generated content should retain provenance such as generated date, model/provider if available and approving instructor.

10. Shared Platform and Data Requirements

10.1 Conceptual Data Domains

Domain	Minimum Records
Identity and access	Users, roles, account states, credentials/provider IDs, permissions and login audit.
Classes	Classes, instructor assignments, learner enrollments, status and module assignments.
Competency content	TESDA source versions, COCs/competencies, modules, scenarios, activities, criteria and rubric weights.
Learning evidence	Attempts, action events, responses, sequence, elapsed time, submission and retry history.
Evaluation	Automated criterion results, provisional score, adjustments, final score, remarks and release state.

Domain	Minimum Records
Gamification	Approved point rules, points, achievements, rewards, levels and progress.
Files	Uploaded resources, metadata, access scope, owner, version and status.
Analytics and reports	Aggregates, exports and report snapshots derived from source records.
Audit	Security, role, access, bypass, content, score and deletion events.

10.2 Shared-Service Requirements

ID	Requirement	Priority	Acceptance Summary
SHR-001	Provide one role-aware authentication and authorization layer for mobile and web.	P0	Unauthorized API access is rejected.
SHR-002	Expose secure APIs/services for classes, content, attempts, evaluation, scores, files, progress and reporting.	P0	Mobile and web reconcile to the same records.
SHR-003	Use stable identifiers and timestamps across all products.	P0	No ambiguous duplicate entities.
SHR-004	Preserve attempt, score and audit history; use archive/deactivation where deletion would destroy required evidence.	P0	Historical reports remain reproducible.
SHR-005	Validate files, size limits, content type and access scope before storage or delivery.	P0	Unauthorized file access is blocked.
SHR-006	Prevent duplicate submission and double scoring using idempotent submission logic or equivalent safeguards.	P0	Repeated requests do not duplicate final records.
SHR-007	Support versioning for modules, criteria and TESDA rubric source so published attempts remain reproducible.	P0	Old attempts retain their original version.
SHR-008	Provide data export and backup capability suitable for defense preparation and recovery.	P0	Export and restore are tested.
SHR-009	Record audit events for all high-impact mutations.	P0	Audit fields identify actor, time, action and target.
SHR-010	Keep implementation compatible with the existing codebase unless a documented migration decision is approved.	P0	No unnecessary rewrite is required.

10.3 Recommended Minimum Entities

Users, Roles/Permissions, Classes, Class Instructors, Enrollments, TESDA Source Versions, Competencies/COCs, Modules, Module Versions, Scenarios, Activities, Criteria/Rubric Items, Files, Quizzes/Quiz Items, Assignments, Attempts, Attempt Events, Automated Evaluations, Score Adjustments, Final Scores, Progress Records, Achievements/Points, Reports and Audit Logs.

This list is conceptual. It must be mapped to the existing database rather than copied blindly as a new schema.

11. Permission Matrix

Action	Learner	Instructor	Admin
Log in to assigned product	Yes	Yes	Yes
View own profile and personal progress	Yes	Own account	Own account
Perform simulations and quizzes	Yes	Preview/test only	No routine use
View learner attempts and criterion evidence	Own only	Assigned classes	System-wide read/audit
Create and publish learning content	No	Yes	Govern/disable; not routine author
Manage classes and enrollments	No	Assigned classes	System-wide governance
Deactivate learner account	No	Assigned learner; reason required	Yes
Delete learner/instructor account	No	No	Admin only
Bypass COC 1-4	No	Yes; audited	Configure permission/audit
Edit and finalize learner score	No	Yes; audited	Emergency-only privilege
Manage roles and access	No	No	Yes
Manage global system settings	No	No	Yes
View audit logs	Own activity only if shown	Own instructional actions	System-wide
Generate reports	Personal only	Assigned scope	System-wide

12. Core Workflows

12.1 Account and Class Access

- Admin creates or imports an account and assigns a role.
- Admin assigns an instructor scope; Instructor creates a class and enrolls learners.
- Learner logs in to mobile; backend verifies role, active state and class membership.
- Assigned modules become visible according to availability and sequence rules.

12.2 Learner Attempt and Automated Evaluation

- Learner opens a published scenario and begins an attempt.
- Mobile records actions and responses while preventing duplicate final submission.
- Evaluation service applies the attached rule/rubric version and creates criterion-level results.
- Provisional feedback is stored; release depends on activity policy and instructor review.

12.3 Instructor Score Review

14. Instructor opens the attempt evidence and automated criterion results.
15. Instructor confirms the score or edits affected criteria/total with required reason and remarks.
16. System stores original and revised values, finalizes the score and records the audit event.
17. Instructor releases the result; mobile shows final status and feedback.

12.4 COC Bypass

18. Instructor selects learner/class and COC sequence to bypass.
19. Instructor supplies a reason and effective scope/date.
20. System records the audit event and unlocks configured content.
21. Bypass does not award completion, points or a passing score.

12.5 Deactivation and Deletion

22. Instructor may deactivate an assigned learner with a reason; records remain intact.
23. Admin may reactivate, deactivate or initiate permanent removal under retention policy.
24. Permanent removal must not break required score, report or audit integrity.
25. All state changes are logged and visible to authorized users.

12.6 Content Publication

26. Instructor drafts module/scenario/activity/quiz and maps it to competency criteria.
27. Instructor previews interaction, scoring and learner view.
28. System validates required rubric mapping and file access before publication.
29. Only the published version is assigned to learners; submitted attempts retain their original version.

13. Non-Functional Requirements

ID / Area	Requirement
NFR-001 Security	Backend-enforced role-based access; secure sessions; protected credentials; least privilege; no learner access to administrative endpoints.
NFR-002 Data integrity	Transactions or equivalent safeguards for attempt submission, scoring and finalization; no partial final score records.
NFR-003 Auditability	High-impact changes record actor, timestamp, old/new state, reason and target.
NFR-004 Performance	Common mobile and dashboard views should become usable within approximately three seconds under normal campus connectivity; scoring feedback should return promptly after submission. Exact targets should be validated on the deployed environment.
NFR-005 Reliability	Graceful error states, retry-safe submissions, no silent data loss and recovery after interrupted connectivity.
NFR-006 Usability	Consistent navigation, readable terminology, clear progress/status labels and minimal visual clutter.
NFR-007 Accessibility	Scalable text, zoom/full-screen, sufficient contrast, dashboard keyboard support and mobile touch targets.
NFR-008 Compatibility	Web supports the browsers used for deployment; mobile supports the platforms already targeted by the existing Flutter project.

ID / Area	Requirement
NFR-009 Maintainability	Typed interfaces, modular services, validation, tests and documented configuration; no unnecessary full rewrite.
NFR-010 Privacy	Only authorized users can view learner records; reports and exports follow institutional privacy practices.
NFR-011 Backup and recovery	Periodic backup, tested restore and export of essential accounts, content, attempts and scores.
NFR-012 Observability	Errors and critical service failures are logged sufficiently to diagnose internal system problems.
NFR-013 Versioning	Published content and rubric changes do not alter historical attempts.
NFR-014 Testability	Rule-based scoring and role permissions can be verified with repeatable automated and manual test cases.

14. Scope Limitations and Explicitly Out-of-Scope Items

14.1 Study-Derived Limitations

- The simulation is limited to a two-dimensional environment.
- Activities use predefined scenarios and cannot represent every real-world CSS situation.
- The study is limited to selected ICT learners of Davao del Norte State College unless the research scope is formally expanded.
- The platform supplements learning and assessment; it does not replace actual hands-on laboratory training.
- Advanced hardware configurations are not part of the baseline scope.
- The platform is not an official TESDA competency assessment, certification or credential-issuing tool.

14.2 Product Boundaries Added to Control Scope

Out of Scope	Boundary
Full LMS replacement	No attendance system, extensive messaging, parent portal, billing, course marketplace or enterprise SIS integration in P0.
3D / AR / VR	No 3D laboratory, augmented reality, virtual reality or physics-accurate hardware emulation.
Autonomous AI assessment	AI does not grade practical competency, finalize scores or publish content without instructor review.
Official TESDA integration	No claim of direct TESDA database integration, official assessor workflow or certificate issuance.
Cross-institution multi-tenancy	No multi-school tenant administration unless separately approved.
Advanced proctoring	No facial recognition, surveillance, browser lockdown or remote examination proctoring.
Real device control	No control of physical computers, routers or laboratory equipment from the mobile app.
Instructor deletion rights	Instructors cannot permanently delete learner or instructor accounts.

Out of Scope	Boundary
Unbounded file storage	Uploads are subject to administrator-configured file types and size limits.
Technology rewrite	No migration away from the confirmed Flutter and Next.js/Node.js/TypeScript codebases without a documented technical reason.

15. Priority and Release Baseline

Priority	Included Work
P0 - Must be completed before capstone acceptance	Study objectives; TESDA source/version mapping; learner mobile simulation; automated rule-based evaluation; real-time progress; separate Instructor/Admin workspaces; classes/enrollment; files; COC bypass; score adjustment/finalization; deactivation/deletion rules; zoom; hidden drag targets; multiple interaction types; security and audit.
P1 - Add after P0 is stable	Instructor-reviewed generative AI quiz drafts; richer video scenarios; more report formats; optional guided practice hints; deeper intervention analytics.
P2 - Future enhancement/research	3D/AR/VR; adaptive machine learning; advanced hardware; multi-institution deployment; official external integrations; advanced proctoring.

15.1 Recommended Implementation Order for the Existing Code

30. Audit the existing Flutter and web code against the requirement IDs in this PRD; do not start by rewriting the UI.
31. Confirm the shared backend, database, authentication provider and current data model.
32. Implement or correct role boundaries and account states before building additional screens.
33. Establish versioned competencies, COCs, modules, criteria and score records.
34. Complete learner attempt capture and the deterministic evaluation pipeline.
35. Complete Instructor class, content, bypass, score review and analytics workflows.
36. Complete Admin account, access, configuration, audit and recovery workflows.
37. Apply panel UI corrections and add non-drag interaction types.
38. Run integrated functional, security, usability and performance tests.
39. Only then evaluate P1 AI quiz generation and other enhancements.

16. Acceptance and Definition of Done

Acceptance Area	Definition of Done
Objective coverage	All five study objectives have at least one implemented, tested and demonstrated requirement.
Panel coverage	All mandatory panel recommendations are mapped to implemented features or explicitly approved limitations.
Role separation	Learner, Instructor and Admin access tests prove that unauthorized actions are blocked at the backend.

Acceptance Area	Definition of Done
TESDA traceability	Every published assessment references an approved source/rubric version and criterion mapping.
Evaluation accuracy	Configured test attempts produce expected criterion results and totals; score edits preserve original values.
Data synchronization	A learner action in mobile appears correctly in the Instructor dashboard and reports without duplicate records.
Historical integrity	Published content/rubric updates do not alter previous attempts or finalized results.
UI readiness	Demonstration supports zoom/full-screen; assessment drag targets are not permanently visible; visual hierarchy is balanced.
Reliability	Interrupted or repeated submission does not create duplicate final attempts or scores.
Documentation	System diagrams, technology descriptions, database documentation and user manuals match the actual implementation.
Testing	Unit, integration, functional, usability, performance and security testing results are recorded for the final system.
Scope compliance	The product clearly states that it supplements training and is not an official TESDA certification tool.

17. Development Gap-Audit Checklist

Instructions. For each item, inspect the existing code and database. Replace “To verify” with Implemented, Partial, Missing or N/A; add the exact file, route, table, API, screenshot or test as evidence. The status is intentionally not pre-filled because the codebase was not reviewed in this task.

Group	ID	Audit Item	Status	Evidence / Next Action
Foundation	FND-01	Shared authentication works for mobile and web.	To verify	
Foundation	FND-02	Backend enforces Learner, Instructor and Admin permissions.	To verify	
Foundation	FND-03	Account active/deactivated/deleted states are modeled.	To verify	
Foundation	FND-04	Mobile and web use the same user/class/content/score records.	To verify	
Foundation	FND-05	Critical actions create audit records.	To verify	
Foundation	FND-06	Backup/export and recovery process exists.	To verify	
Mobile	MOB-01	Learner-only mobile navigation and role validation.	To verify	
Mobile	MOB-02	Class and assignment display.	To verify	

Group	ID	Audit Item	Status	Evidence / Next Action
Mobile	MOB-03	COC/module sequence and lock states.	To verify	
Mobile	MOB-04	Text and video scenario support.	To verify	
Mobile	MOB-05	At least three interaction types beyond static content.	To verify	
Mobile	MOB-06	Drag-and-drop assessment has no permanent target boxes.	To verify	
Mobile	MOB-07	Zoom and full-screen support.	To verify	
Mobile	MOB-08	Attempt event/evidence capture.	To verify	
Mobile	MOB-09	Automated criterion feedback and provisional score.	To verify	
Mobile	MOB-10	Final score release state.	To verify	
Mobile	MOB-11	Progress, achievements and points.	To verify	
Mobile	MOB-12	Retry rules preserve history.	To verify	
Mobile	MOB-13	Duplicate submission protection.	To verify	
Mobile	MOB-14	Balanced visual hierarchy and minimized logos.	To verify	
Instructor	INS-01	Separate Instructor workspace and route guards.	To verify	
Instructor	INS-02	Class create/edit/archive.	To verify	
Instructor	INS-03	Enrollment and class membership management.	To verify	
Instructor	INS-04	Instructor can deactivate but not delete assigned learners.	To verify	
Instructor	INS-05	Module/scenario/activity/quiz CRUD and publishing.	To verify	
Instructor	INS-06	File upload and access control.	To verify	
Instructor	INS-07	TESDA rubric/criteria mapping.	To verify	
Instructor	INS-08	COC 1-4 bypass with reason and audit.	To verify	
Instructor	INS-09	Class and learner analytics.	To verify	
Instructor	INS-10	Attempt evidence review.	To verify	
Instructor	INS-11	Score adjustment with reason and original value.	To verify	
Instructor	INS-12	Finalization and learner release.	To verify	
Instructor	INS-13	Report generation/export.	To verify	
Instructor	INS-14	Instructor scope limits cross-class data access.	To verify	
Admin	ADM-01	Separate Admin workspace and route guards.	To verify	

Group	ID	Audit Item	Status	Evidence / Next Action
Admin	ADM-02	Account and role management.	To verify	
Admin	ADM-03	Admin-only permanent deletion with safeguards.	To verify	
Admin	ADM-04	Instructor scope/permission management.	To verify	
Admin	ADM-05	Global system configuration.	To verify	
Admin	ADM-06	TESDA source/rubric version activation.	To verify	
Admin	ADM-07	System-wide audit log.	To verify	
Admin	ADM-08	Storage/file governance.	To verify	
Admin	ADM-09	System analytics and reports.	To verify	
Admin	ADM-10	Emergency score correction is exceptional and audited.	To verify	
Evaluation and Data	EVA-01	Criteria-level deterministic rules.	To verify	
Evaluation and Data	EVA-02	Rubric version remains linked to historical attempts.	To verify	
Evaluation and Data	EVA-03	Competency score is separate from gamification points.	To verify	
Evaluation and Data	EVA-04	Provisional and final states are distinct.	To verify	
Evaluation and Data	EVA-05	Score revisions preserve history.	To verify	
Evaluation and Data	EVA-06	COC bypass does not auto-pass or auto-award points.	To verify	
Evaluation and Data	EVA-07	Reports reconcile with source attempts and final scores.	To verify	
Evaluation and Data	EVA-08	File and report access follows role/class scope.	To verify	
Quality	QAT-01	Unit tests cover scoring and core services.	To verify	
Quality	QAT-02	Integration tests cover mobile-web-data flow.	To verify	
Quality	QAT-03	Authorization/security tests pass.	To verify	
Quality	QAT-04	Usability test covers learners and instructors.	To verify	
Quality	QAT-05	Performance is acceptable on deployment devices/network.	To verify	
Quality	QAT-06	Documentation matches Flutter/Next.js/Node.js/TypeScript implementation.	To verify	

Appendix A. Panel Recommendation Traceability

No.	Panel Recommendation	Mapped Requirement	Priority
1	Grading and modules based on TESDA booklet	ADM-007, INS-007, EVAL-001 to EVAL-010	P0
2	Separate Instructor system; Admin handles internal, Instructor external	Section 4; INS-001; ADM-001	P0
3	Instructor can bypass COC 1-4	INS-010; EVAL-007; Workflow 12.4	P0
4	Zoom screen during demonstration	MOB-007; UI-001	P0
5	Remove boxes in drag-and-drop	MOB-006; UI-002	P0
6	Points based on TESDA standards	Section 8.1; EVAL-001/008	P0
7	Instructor can edit/check score and record final score	INS-012/013/014; EVAL-005/006	P0
8	Use AI if suitable; generate instructor quizzes	INS-008; Section 9.3	P1
9	Instructor classes/LMS enrollment; Admin manages access	INS-002/003; ADM-002/004	P0
10	Instructor manages/uploads files and monitors students	INS-006/011/015	P0
11	Instructor deactivates; only Admin removes users	INS-004; ADM-003	P0
12	Scenario activities may use video or text	MOB-004; Section 9.1	P0
13	More interactive than drag-and-drop	MOB-005; UI-003	P0
14	Minimize logos and figures	MOB-014; UI-005	P0

Appendix B. Open Decisions

ID	Decision Required	Owner
OD-01	Which exact TESDA booklet/training regulation edition is approved as the scoring and module source of truth?	Study team + Instructor
OD-02	What are the exact COC 1-4 names, sequence, pass thresholds and task weights?	Instructor + research adviser
OD-03	What backend, database, authentication and file-storage providers are already used by the codebases?	Developers
OD-04	Which Flutter deployment targets are included in the capstone release?	Developers + adviser
OD-05	What file types, maximum sizes and retention periods are allowed?	Admin + adviser
OD-06	Can an Instructor reactivate a learner they deactivated, or is reactivation Admin-only?	Adviser + policy owner

ID	Decision Required	Owner
OD-07	Which reports and export formats are required for defense and final users?	Instructor + panel expectations
OD-08	Are automated scores always reviewed, or only flagged/assessment attempts?	Instructor + adviser
OD-09	Which activities show immediate feedback and which withhold feedback until completion?	Instructor
OD-10	Which AI provider, privacy rules and cost limits apply if P1 quiz generation is approved?	Admin + developers
OD-11	What conditions permit permanent deletion while preserving required academic and audit history?	Admin + institutional policy
OD-12	What data is included in real-time analytics versus scheduled aggregates?	Developers + instructor

No-guess rule

Development should not invent the missing TESDA weights, thresholds or institutional policies. Resolve and document each open decision before coding the affected rule.

Appendix C. Glossary

Term	Definition
Automated skill evaluation	Rule-based and score-based processing of learner evidence against predefined competency criteria.
COC	A competency grouping used by the project and panel recommendations. Exact names and sequence must follow the approved TESDA source.
Competency score	Assessment-oriented score based on approved criteria and task weights.
Gamification points	Engagement/reward points that remain distinct from the competency score.
Provisional score	Automated result that may still require instructor review.
Final score	Instructor-confirmed or adjusted result recorded as the authoritative system score for the learning activity.
Bypass	Audited permission to access a later COC/module without automatically granting completion or competency.
Internal problem	Account, access, configuration, storage, security or platform concern handled by Admin.
External problem	Class, content, learning, attempt or scoring concern handled by Instructor.
LMS-lite	Limited class, enrollment, assignment, content and monitoring features needed for this capstone, not a full enterprise LMS.

Term	Definition
Rubric version	Immutable version of competency criteria and weights attached to a published assessment and its attempts.
Supplementary tool	A practice and monitoring system that supports but does not replace hands-on training or official assessment.

Source Basis Note

This PRD is grounded in the uploaded “A Gamified Simulation Platform with Automated Skill Evaluation for NC II Computer Systems Servicing” Chapter 1-2 draft. Key source areas reviewed include Objectives of the Study (pp. 3-4), Scope and Limitation (p. 5), use cases and system boundaries (pp. 18-24), rule-based evaluation and score logic (pp. 35-40), analytics (p. 41), and testing/security expectations (pp. 41-44). The fourteen outline-defense comments supplied by the proponents were treated as additional product requirements. No exact TESDA scoring values were introduced because the booklet itself was not provided in this task.

END OF PRODUCT REQUIREMENTS DOCUMENT